MATHS MANTHAN
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 04
Chapter 13: Statistics

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Weekly Test 04

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. In the formula Mode $= l + \left(\dfrac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$, the symbol f_0 represents: [1]

(A) frequency of the modal class
(B) frequency of the class preceding the modal class
(C) frequency of the class succeeding the modal class
(D) cumulative frequency of the modal class

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) frequency of the class preceding the modal class
Justification: In the standard CBSE mode formula, f_1 is the frequency of the modal class, f_0 is the frequency of the class preceding the modal class, and f_2 is the frequency of the class succeeding the modal class.

Q2. The construction of a cumulative frequency table is useful in determining the: [1]

(A) Mean (B) Median (C) Mode (D) All of these

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) Median
Justification: Cumulative frequency identifies the position of the middle observation ($\frac{N}{2}$) and the corresponding median class; hence, cumulative frequency tables are specifically constructed to find the median.

Q3. If each observation of a statistical data having mean $\bar{x}$ is multiplied by k ($k \neq 0$) and then increased by c, the new mean is: [1]

(A) $k\bar{x}$ (B) $\bar{x} + c$ (C) $k\bar{x} + c$ (D) $k(\bar{x} + c)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $k\bar{x} + c$

Justification: Let original observations be x_i with mean $\bar{x} = \frac{1}{n}\sum x_i$.
New observations $y_i = kx_i + c$.

$$\bar{y} = \frac{\sum(kx_i + c)}{n} = \frac{k\sum x_i + nc}{n} = k\left(\frac{\sum x_i}{n}\right) + c = \mathbf{k\bar{x} + c}.$$

$$\boxed{\text{New Mean} = \mathbf{k\bar{x} + c}}$$

Q4. For the distribution: [1]

Marks	Below 10	Below 20	Below 30	Below 40	Below 50
Students	3	12	27	57	75

The modal class is:

(A) $10 - 20$ (B) $20 - 30$ (C) $30 - 40$ (D) $40 - 50$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $30 - 40$

Justification: Converting the cumulative "Below" frequencies into regular class frequencies:

- $0 - 10$: 3
- $10 - 20$: $12 - 3 = 9$
- $20 - 30$: $27 - 12 = 15$
- $30 - 40$: $57 - 27 = \mathbf{30}$ (Maximum Frequency)
- $40 - 50$: $75 - 57 = 18$

The maximum frequency is 30, which corresponds to the class interval $\mathbf{30 - 40}$.

Q5. **Assertion (A):** The modal class of a grouped frequency distribution is the class having the highest frequency.
Reason (R): Mode is the value of the observation having maximum frequency in a data set. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: By definition, mode represents the observation with the highest frequency. In a grouped frequency distribution, the interval having the greatest frequency is defined as the modal class. Therefore, both (A) and (R) are true, and (R) provides the correct explanation of (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. Find the mode of the following set of observations:
$14, 25, 14, 28, 18, 17, 18, 14, 23, 22, 14, 18.$ [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Step 1: Frequency Tabulation of Observations [1 Mark]

Listing the distinct values and their respective frequencies:

Observation	14	17	18	22	23	25	28
Frequency	**4**	1	3	1	1	1	1

Step 2: Identifying the Mode [1 Mark]

The observation 14 occurs maximum number of times (frequency = 4).

$$\boxed{\text{Mode of the observations} = \mathbf{14}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q7. Find the missing frequency f if the mode of the following distribution is 65: [3]

Class	20 – 40	40 – 60	60 – 80	80 – 100	100 – 120
Frequency	10	f	25	16	10

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Modal Class & Parameter Identification [1 Mark]

Given Mode = 65, which lies in the class interval **60 – 80**.
$\therefore$ Modal Class = 60 – 80.
Lower limit $(l) = 60$, class width $(h) = 20$.
$f_1 = 25$ (modal class frequency), $f_0 = f$ (preceding frequency), $f_2 = 16$ (succeeding frequency).

Step 2: Mode Formula & Equation Formulation [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$65 = 60 + \left(\frac{25 - f}{2(25) - f - 16}\right) \times 20 \implies 5 = \frac{20(25 - f)}{34 - f}$$

Step 3: Solving for f & Final Boxed Answer [1 Mark]

Dividing both sides by 5:

$$1 = \frac{4(25 - f)}{34 - f} \implies 34 - f = 100 - 4f$$

$$4f - f = 100 - 34 \implies 3f = 66 \implies f = 22.$$

$$\boxed{\text{Missing Frequency } f = \mathbf{22}}$$

Q8. Calculate the median marks for 50 students from the following data: **[3]**

Marks	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$
Students	6	11	17	12	4

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Cumulative Frequency Table [1 Mark]

Marks	Frequency (f_i)	Cumulative Frequency (cf)
$0-10$	6	6
$10-20$	11	$6+11=17$
$20-30$	17	$17+17=34$
$30-40$	12	$34+12=46$
$40-50$	4	$46+4=50$
Total	$N=\sum f_i=50$	

Step 2: Median Class & Parameters [1 Mark]

$N=50 \implies \dfrac{N}{2}=25$.

Cumulative frequency just greater than 25 is 34, which corresponds to the class **$20-30$**.

$\therefore$ Median Class $=20-30$.

Lower limit $(l)=20$, cf of preceding class $=17$, frequency $(f)=17$, class width $(h)=10$.

Step 3: Median Formula & Calculation [1 Mark]

$$\text{Median}=l+\left(\frac{\frac{N}{2}-cf}{f}\right)\times h=20+\left(\frac{25-17}{17}\right)\times 10=20+\frac{80}{17}$$

$$\text{Median}=20+4.706\approx 24.71 \quad \left(\text{or } 24\frac{12}{17}\right)$$

$$\boxed{\text{Median Marks}=\mathbf{24.71} \quad \left(\text{or } \mathbf{24\frac{12}{17}}\right)}$$

Q9. Find the arithmetic mean of the distribution using an appropriate method: **[3]**

Class	$100-120$	$120-140$	$140-160$	$160-180$	$180-200$
Frequency	12	14	8	6	10

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Step-Deviation Calculation Table [1.5 Marks]

Let Assumed Mean $a=150$, class width $h=20$, $u_i=\dfrac{x_i-150}{20}$.

Class	Frequency (f_i)	Class Mark (x_i)	$u_i=\frac{x_i-150}{20}$	f_iu_i
$100-120$	12	110	-2	-24
$120-140$	14	130	-1	-14
$140-160$	8	150	0	0
$160-180$	6	170	$+1$	$+6$
$180-200$	10	190	$+2$	$+20$
Total	$\sum f_i=50$			$\sum f_iu_i=-12$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x}=a+\left(\frac{\sum f_iu_i}{\sum f_i}\right)\times h=150+\left(\frac{-12}{50}\right)\times 20=150-\frac{240}{50}=150-4.8=145.2$$

Step 3: Final Boxed Result [0.5 Mark]

$$\boxed{\text{Arithmetic Mean }(\bar{x})=\mathbf{145.2}}$$

SECTION D: Case Study Based Question

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Q10. CASE STUDY: VOCATIONAL TRAINING INSTITUTE YOUTH SKILL DEMOGRAPHICS [4 Marks]

A state vocational technical council analyzed the age profiles of students and young adults enrolled in digital manufacturing and industrial robotics certification courses.

Age Group (years)	$15-20$	$20-25$	$25-30$	$30-35$	$35-40$	$40-45$
Enrolled Candidates	60	110	80	50	40	20

(i) What is the total number of candidates enrolled and which class has highest density? [1]

(ii) Calculate the modal age of the candidates undergoing training. [2]

(iii) Find the median class interval for the dataset. [1]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Subpart (i): Total Candidates & Highest Density Class [1 Mark]

$$\text{Total Candidates } N = 60 + 110 + 80 + 50 + 40 + 20 = \mathbf{360}.$$

The class with the highest frequency (110) is $\mathbf{20-25}$ years.

$$\boxed{\text{Total Candidates} = \mathbf{360}, \quad \text{Highest Density Class} = \mathbf{20-25} \text{ years}}$$

Subpart (ii): Modal Age Calculation [2 Marks]

For modal class $20-25$: $l = 20$, $h = 5$, $f_1 = 110$, $f_0 = 60$, $f_2 = 80$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \text{ [1 Mark]}$$

$$\text{Mode} = 20 + \left(\frac{110-60}{2(110)-60-80}\right) \times 5 = 20 + \left(\frac{50}{220-140}\right) \times 5 = 20 + \left(\frac{50}{80}\right) \times 5$$

$$\text{Mode} = 20 + \frac{250}{80} = 20 + 3.125 = \mathbf{23.125} \text{ years} \quad \left(\text{or } 23\frac{1}{8} \text{ years}\right) \text{ [1 Mark]}$$

$$\boxed{\text{Modal Age} = \mathbf{23.125} \text{ years} \quad \left(\text{or } \mathbf{23\frac{1}{8}} \text{ years}\right)}$$

Subpart (iii): Median Class Interval [1 Mark]

$N = 360 \implies \dfrac{N}{2} = 180.$

Cumulative frequencies: $15-20 : 60$; $20-25 : 170$; $25-30 : 250$; $30-35 : 300$; $35-40 : 340$; $40-45 : 360$.

The cumulative frequency just greater than 180 is 250, corresponding to class $\mathbf{25-30}$.

$$\boxed{\text{Median Class Interval} = \mathbf{25-30} \text{ years}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★